SMART POWER MANAGEMENT

Research Paper Summaries

This document summarizes the three research papers provided. Each paper is explained in simple language, covering its aim, approach, important findings, and relevance to a Smart Power Management project.

Research papers included

• 1. Reduce Cost Smart Power Management System by Utilize Single Board Computer Artificial Neural Networks for Smart Systems (2019)

• 2. A Comprehensive Review of Smart Energy Management Systems for Photovoltaic Power Generation Utilizing the Internet of Things (2025)

• 3. Design, Implementation, and Deployment of an IoT Based Smart Energy Management System (IEEE Access; DOI dated 2021)

Note: Findings and numbers below are reported by the papers; they are not a guarantee of performance for every real-world installation.

Research Paper 1

Reduce Cost Smart Power Management System by Utilize Single Board Computer Artificial Neural Networks for Smart Systems

Authors: Sudad J. Ashaj and Ergun Erçelebi | Journal: International Journal of Computational Intelligence Systems, 2019
DOI: https://doi.org/10.2991/ijcis.d.191001.001

1. Aim of the Research

The paper proposes an affordable smart power management system for homes and buildings. Its purpose is to reduce unnecessary electricity use by automatically controlling lights, a television, air conditioning, and ventilation based on room occupancy, human activity, temperature, and gas or odour readings.

2. Method / System Approach

The system uses a Raspberry Pi 3 Model B as the main controller. A camera captures video, while light, temperature/humidity, ultrasonic-distance, gas, and power sensors provide additional information. Python software processes the inputs. A convolutional neural network (CNN), combined with saliency-based object detection, identifies people and objects such as chairs or sofas. Relays then switch appliances on or off according to the system's decisions.

3. Key Findings

The researchers tested the setup in an apartment containing rooms and a hall. The system automatically controlled lights and other connected devices during the tests. The paper reports human/object detection around 99–100% in its experiments, with recognition response times around 0.9 seconds; the abstract also reports detection in about 0.7 seconds. The authors note that ultrasonic distance measurements had some errors.

4. Relevance to the Project

This paper is useful for a Smart Power Management project that switches appliances based on occupancy instead of leaving them running unnecessarily. It demonstrates how a low-cost single-board computer, sensors, computer vision, and relays can work together. A practical project should also consider privacy, electrical safety, and testing in different rooms.

Keywords: Raspberry Pi, CNN, sensors, occupancy detection, automatic appliance control, energy saving.

Research Paper 2

A Comprehensive Review of Smart Energy Management Systems for Photovoltaic Power Generation Utilizing the Internet of Things

Authors: Challa Krishna Rao, Sarat Kumar Sahoo, and Franco Fernando Yanine | Journal: Unconventional Resources, 2025
DOI: https://doi.org/10.1016/j.uncres.2025.100197

1. Aim of the Research

This review examines how the Internet of Things (IoT) can help monitor, manage, and improve solar photovoltaic (PV) power systems. It brings together earlier research on sensors, communications, cloud computing, forecasting, automated control, and the integration of solar power with the electricity grid.

2. Method / System Approach

Rather than presenting one new hardware prototype, the authors review and compare existing research and system designs. Typical systems collect solar-panel voltage, current, temperature, and energy-output data using sensors and smart meters. Data can be sent through Wi-Fi, GSM, LoRa, or other networks to cloud platforms for dashboards and analysis. Controllers can then support load balancing, battery charging, grid interaction, fault alerts, and predictive maintenance.

3. Key Findings

The review describes AI and machine learning for solar-output forecasting and early fault detection, edge/fog computing for faster local processing, smart storage and demand response, and newer connectivity options such as 5G and LPWAN. It discusses possible benefits including improved monitoring, more efficient use of generated power, and reduced operating costs. The paper also highlights barriers such as cybersecurity and privacy risks, initial cost, unreliable internet, interoperability, data management, and system scalability. Its reported efficiency and savings ranges summarize reviewed studies and should not be treated as guaranteed results for every system.

4. Relevance to the Project

This paper provides background for adding IoT and solar monitoring to a Smart Power Management project. It can help justify features such as live energy readings, remote dashboards, fault alerts, and future solar/battery integration. It also reminds project designers to plan for security, connectivity failures, and compatibility between devices.

Keywords: IoT, solar PV, cloud computing, forecasting, energy monitoring, smart grid.

Research Paper 3

Design, Implementation, and Deployment of an IoT Based Smart Energy Management System

Authors: M. Usman Saleem, M. Rehan Usman, and Mustafa Shakir | IEEE Access
DOI: https://doi.org/10.1109/ACCESS.2021.3070960

1. Aim of the Research

The paper designs and tests an IoT-based Smart Energy Management System (SEMS) that helps consumers and facility operators understand electricity use more accurately. It focuses on collecting power data from smart meters, sending it to a central software service, and making usage information available for monitoring and analysis.

2. Method / System Approach

The proposed system uses smart meters to measure electrical parameters such as voltage, current, active/reactive power, energy consumption, and power factor. Microcontrollers and a NodeMCU/Wi-Fi connection transmit the readings to a middleware module and cloud-based service. The Entrack software provides views of real-time, daily, and monthly consumption. The system was installed and evaluated at four facilities belonging to Stylo Pvt. Ltd. in Lahore, Pakistan.

3. Key Findings

The case study showed that the system could collect and display consumption data at different time scales and across different facility types. The authors explain that seeing real-time and historical usage can help users identify unusual consumption, conduct energy audits, and reduce waste. They also describe demand-response actions, such as generating an overload notification and allowing non-essential loads to be disabled. The paper's main emphasis is monitoring and data visibility; expanded internet-based control is described as further work.

4. Relevance to the Project

This paper is a strong reference for a project that monitors electricity consumption and displays readings on a web or mobile dashboard. It shows a practical IoT data path from smart meter to network, middleware, and user interface. A student prototype could implement a smaller version with a suitable sensor/meter module, microcontroller, and dashboard, while following electrical isolation and safety requirements.

Keywords: IoT, smart meter, smart grid, energy monitoring, cloud software, demand-side management.